

Portfolio Risk Analytics & Stress Testing Engine

A Quantitative Risk Report with a Solvency II-Style Insurance Perspective

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Executive Summary

This report analyses the risk profile of a 60/40 equity-bond portfolio built from two liquid, publicly traded ETFs used as market proxies: EUNL.DE (global equities) and EUN5.DE (euro investment-grade corporate bonds). The objective is to demonstrate a practical, multi-method approach to portfolio risk measurement, in a style consistent with risk analysis practices in banking, insurance, and reinsurance.

The portfolio shows an annualized volatility of 10.9%, with a 1.1 percentage point diversification benefit relative to its individually weighted components. For a one-day risk horizon, the 95% historical Value at Risk (VaR) points to a potential loss of approximately 1.0% (€9,900 on a €1,000,000 portfolio), rising to 2.3% (€23,200) at the 99% level. A formal Kupiec backtest does not reject correct coverage of this VaR model at either confidence level — this specifically tests VaR breach frequency, not Expected Shortfall or overall tail accuracy.

The gap between historical VaR and the Normal-distribution-based (parametric and Monte Carlo) estimates widens sharply at the 99% level. An Extreme Value Theory analysis provides further evidence of heavy-tailed risk, suggesting this gap reflects a real feature of the return distribution rather than sampling noise. Historical stress testing identifies the COVID-19 crash as the most severe observed scenario (–23.5%), while the 2022 stock-bond selloff shows that bonds can lose more than equities when both fall together during a rate-hiking period. A duration-based analysis and an illustrative Solvency II-style capital perspective (not a full regulatory SCR calculation) connect these findings to an insurance risk-management context.

No single measure tells the full story on its own — combining them is what gives a realistic picture of both routine and extreme portfolio risk.

1. Project Objective

The goal of this project is to build a Python-based quantitative risk analytics framework using real, public market data, and to apply it to a simple illustrative portfolio. The analysis combines standard market risk techniques (volatility, correlation, Value at Risk, Expected Shortfall, Monte Carlo simulation, stress testing) with two extensions drawn from insurance and actuarial risk management: an interest rate duration shock relevant to asset-liability management, and Extreme Value Theory, used in practice to model catastrophic and tail-risk events.

The portfolio and its allocation are illustrative and do not represent the actual holdings of any institution.

2. Data & Methodology

2.1 Data

Daily closing prices were obtained from Yahoo Finance for two ETFs over the period January 2020 to September 2026 (1,694 trading days after cleaning):

- EUNL.DE — iShares Core MSCI World UCITS ETF, used as a global equity proxy
- EUN5.DE — iShares Core € Corp Bond UCITS ETF, used as a euro investment-grade corporate bond proxy

2.2 Portfolio Construction

A constant-weight 60/40 (equity/bond) allocation is used as a transparent analytical baseline. Note that computing portfolio returns as a fixed-weight combination each day is closer to a daily-rebalanced portfolio than a true buy-and-hold position, whose weights would drift with market prices — a distinction discussed further in Section 9.

2.3 Risk Measures Applied

- Historical, Parametric (Variance-Covariance), and Monte Carlo Value at Risk (95% and 99%)
- Expected Shortfall (average loss beyond the VaR threshold)
- Kupiec Proportion of Failures backtest, to test whether the VaR model's breach rate matches its confidence level
- Historical stress testing (COVID-19 crash, 2022 rate shock, 2022 stock-bond selloff, 2025 tariff shock) and hypothetical single-day shock scenarios
- Correlation with broad market factors: S&P 500, VIX, US 10-Year Treasury yield, EUR/USD
- Interest rate duration shock on the bond position
- An illustrative one-year risk perspective (Solvency II-style framing, not a regulatory SCR calculation)
- Extreme Value Theory (Peaks-Over-Threshold, Generalized Pareto Distribution)

3. Portfolio Risk Characteristics

The two assets show markedly different risk profiles. EUNL.DE has an annualized volatility of 16.5% and a maximum historical drawdown of -33.6%, while EUN5.DE is considerably more stable, at 5.4% annualized volatility and a -17.3% maximum drawdown. The correlation between the two assets over the sample period is 0.40.

Combining them in a 60/40 allocation produces a portfolio annualized volatility of 10.9%, compared with a weighted average of 12.0% for the two assets held separately — a diversification benefit of 1.1 percentage points. The portfolio's own maximum drawdown over the sample period is -24.8%.

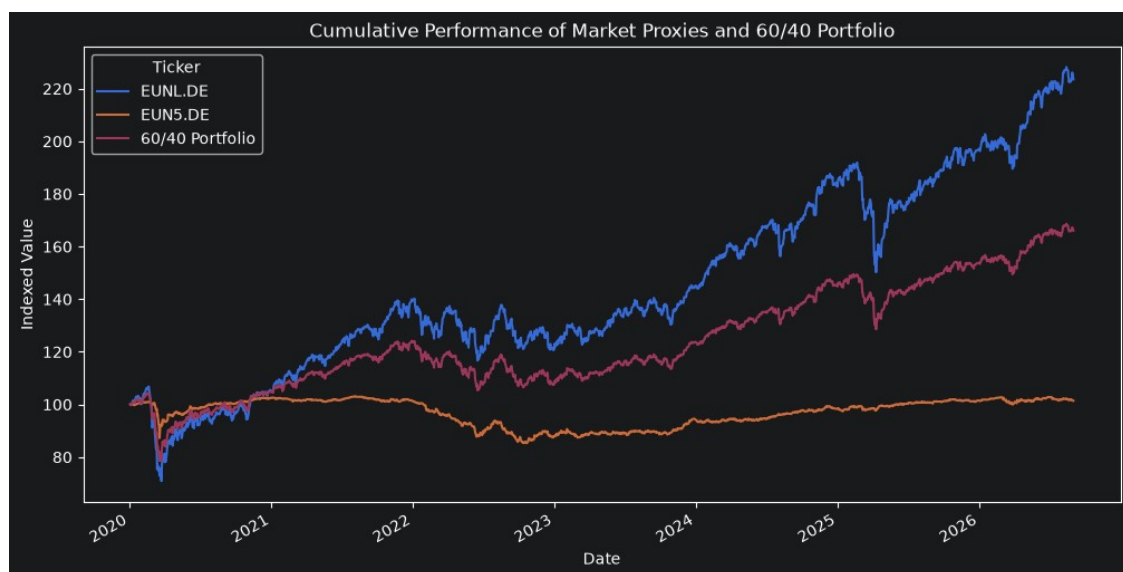


Figure 1. Cumulative performance of the individual market proxies and the 60/40 portfolio, indexed to 100 at the start of the sample period.

Metric	EUNL.DE (Equity)	EUN5.DE (Bond)	60/40 Portfolio
Annualized volatility	16.48%	5.37%	10.93%
Maximum drawdown	-33.63%	-17.31%	-24.77%

Table 1. Volatility and maximum drawdown, individual assets vs. the 60/40 portfolio.

4. Value at Risk & Expected Shortfall

Value at Risk was estimated using three complementary methods: a direct historical simulation, a parametric (Variance-Covariance) approach assuming normally distributed returns, and a Monte Carlo simulation. Expected Shortfall (the average loss beyond the VaR threshold) was calculated to provide additional insight into the severity of losses in the tail.

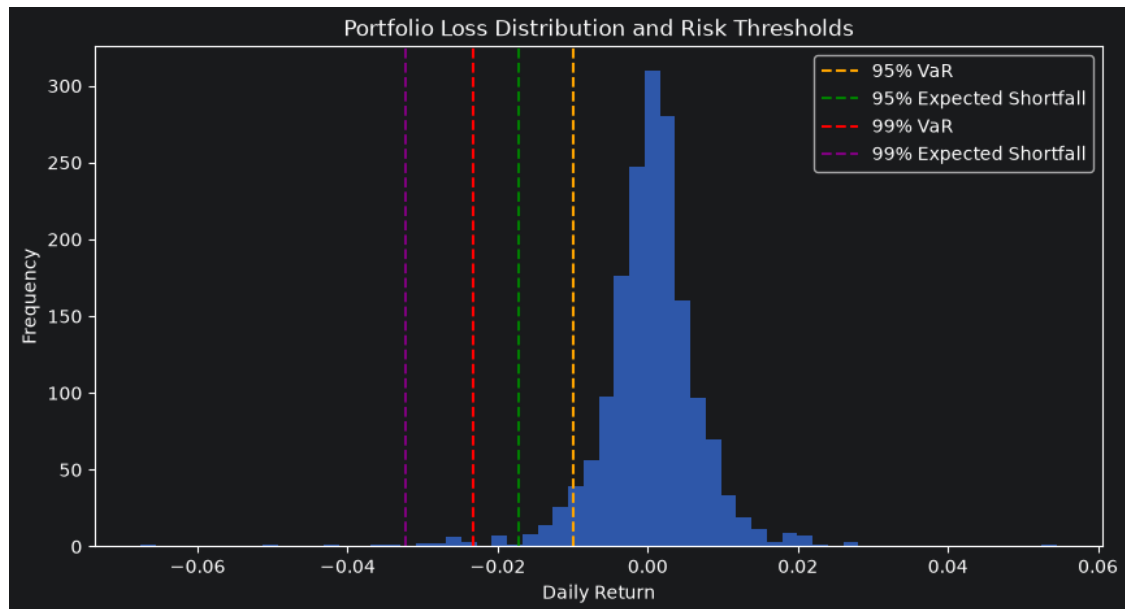


Figure 2. Distribution of daily portfolio returns, with 95% and 99% VaR and Expected Shortfall thresholds.

Measure	95% Confidence	99% Confidence
Historical VaR	-0.99%	-2.32%
Parametric VaR	-1.10%	-1.57%
Monte Carlo VaR	-1.11%	-1.57%
Historical Expected Shortfall	-1.71%	-3.23%

Table 2. Value at Risk by estimation method, and historical Expected Shortfall.

At the 95% level, all three VaR methods are closely aligned. At the 99% level, however, the historical VaR (-2.32%) is clearly worse than the parametric and Monte Carlo estimates (-1.57%) — an early sign that the true return distribution has fatter tails than a Normal distribution would suggest, examined further in Section 8.

4.1 Model Validation: Kupiec Backtest

Actual portfolio losses exceeded the 95% VaR threshold on 85 of 1,694 trading days ($\approx 5.02\%$), and the 99% VaR threshold on 17 occasions ($\approx 1.00\%$) — both close to their theoretical expectations. A formal Kupiec Proportion of Failures test does not reject the null hypothesis of correct unconditional coverage at either confidence level, with p-values of 0.973 (95%) and 0.988 (99%), well above the conventional 5% significance threshold. This means the test finds no statistical evidence against the model; it does not, by itself, prove the model is accurate, and it evaluates VaR breach frequency only — not Expected Shortfall or tail severity.

5. Stress Testing

Stress testing evaluates portfolio behaviour under both real historical market disruptions and hypothetical shocks, to complement the statistical VaR/ES framework with scenario-based analysis.

5.1 Historical Scenarios

Scenario	Equity Return	Bond Return	Portfolio Return
COVID-19 Crash	-33.00%	-9.13%	-23.45%
COVID Crash + Partial Recovery Window	-15.71%	-3.63%	-10.88%
2022 Rate & Inflation Shock	-16.48%	-13.21%	-15.17%
2022 Stock-Bond Selloff	-13.39%	-15.77%	-14.34%
2025 Tariff Shock	-13.38%	-0.93%	-8.40%

Table 3. Portfolio impact across real historical stress periods.

The COVID-19 crash remains the most severe observed scenario, driven primarily by the equity component. During the 2022 stock-bond selloff, the bond position actually lost more than the equity position (–15.8% vs. –13.4%) — a reminder that the diversification benefit of fixed income can weaken or reverse when bond and equity prices fall together, as happened during that period of rising rates. The more recent 2025 tariff shock produced a comparatively milder –8.4% impact.

5.2 Hypothetical Scenarios

Hypothetical scenarios scale the single worst observed daily return for each asset by a multiplier, to illustrate moderate, severe, and extreme single-day shocks: –7.7%, –11.6%, and –15.5% respectively. These are single-day shocks and are not directly comparable, in time horizon, to the multi-week historical scenarios above — only in magnitude.

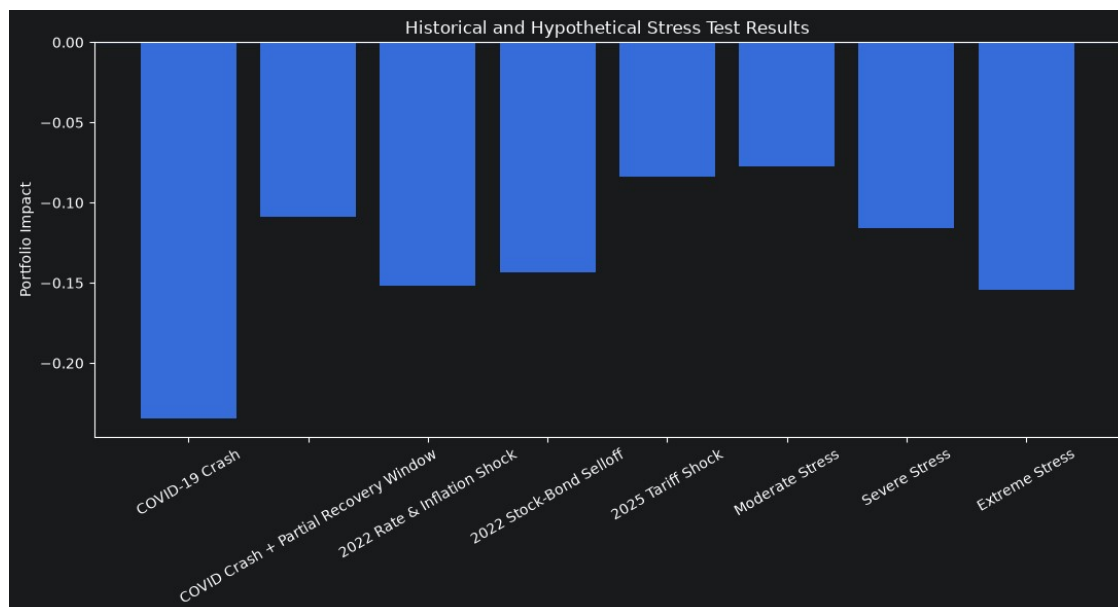


Figure 3. Portfolio impact across all historical and hypothetical stress scenarios.

6. Market Factor Analysis

To understand what drives portfolio risk beyond the two holdings themselves, portfolio returns were compared against four broader market factors: the S&P 500, the VIX volatility index, the US 10-Year Treasury yield, and the EUR/USD exchange rate.

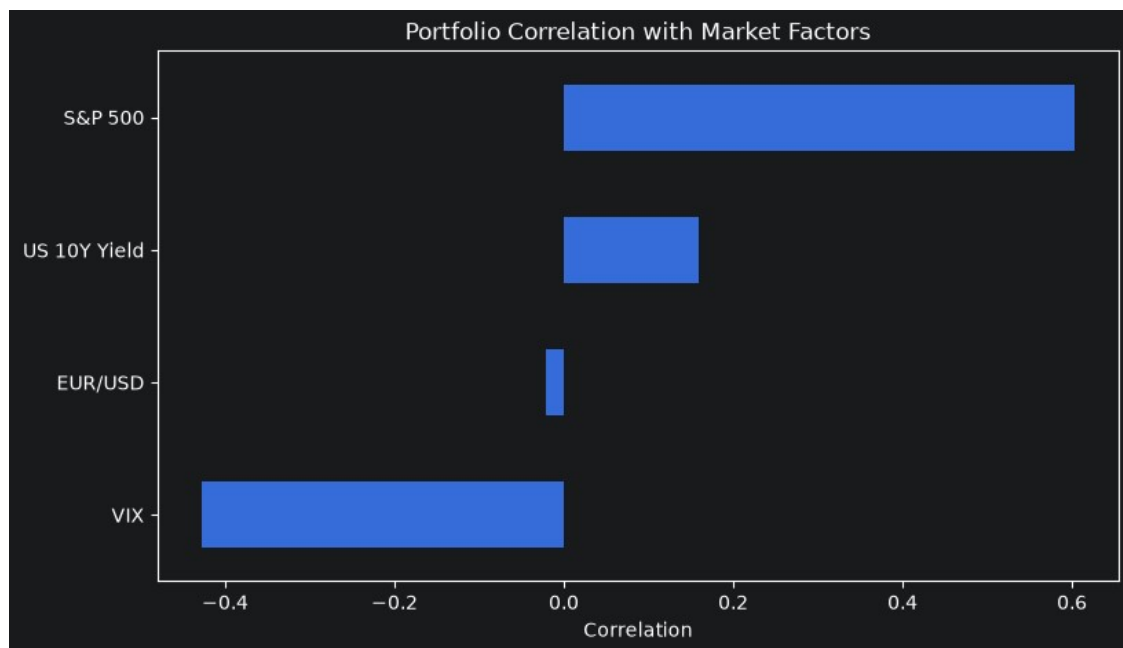


Figure 4. Correlation between portfolio returns and broader market factors.

The portfolio shows a strong positive correlation with the S&P 500 (+0.60), consistent with its equity-heavy composition, and a strong negative correlation with the VIX (-0.43), reflecting its exposure to market fear and volatility. Correlation with the US 10-Year yield (+0.16) and EUR/USD (-0.02) is comparatively weak.

Based on the estimated linear sensitivity (beta) of the portfolio to VIX changes (-0.0362), and holding this estimated relationship constant, a 50% VIX spike corresponds to an illustrative portfolio impact of approximately -1.8%, a 100% spike to -3.6%, and a 200% spike to -7.2%. These are extrapolations of a historical linear relationship, not independent stress scenarios.

7. Interest Rate Risk & Illustrative Capital Perspective

7.1 Interest Rate Duration Shock

Insurance and reinsurance companies are highly sensitive to interest rate movements, given the long-term nature of their liabilities. Using an approximate modified duration of 5.5 years for the bond position (typical of a short-to-medium-term euro corporate bond fund), the estimated impact of a parallel interest rate shift is:

Rate Shock	Bond Price Impact	Portfolio Impact
+50 bps	-2.75%	-1.10%
+100 bps	-5.50%	-2.20%
+200 bps	-11.00%	-4.40%

Table 4. Estimated impact of parallel interest rate shocks.

7.2 One-Year Risk Perspective (Illustrative, Not a Regulatory SCR)

This section connects the portfolio's daily risk estimate to a one-year view, in three steps.

A) Daily statistical estimate. The 99.5% daily historical VaR of the portfolio is -2.74% .

B) Illustrative one-year scaling. Applying a square-root-of-time scaling ($\times\sqrt{252} \approx \times 15.87$) gives an illustrative one-year figure of approximately -43.45% of portfolio value — around €434,000 on a €1,000,000 portfolio. This scaling assumes daily returns are independent and identically distributed, which real markets do not fully satisfy (for example, volatility tends to cluster rather than occur independently day to day).

C) Why this is not the regulatory SCR. Solvency II's actual Solvency Capital Requirement is not a time-scaled VaR of a single portfolio return series. Under the Standard Formula, each risk type (equity, interest rate, spread, currency, and others) is shocked separately according to regulator-defined calibrations, and the resulting sub-module capital charges are combined using prescribed correlation matrices — or, alternatively, derived from a supervisor-approved internal model. The figure above is a simplified statistical illustration meant to connect this project's risk measures to a regulatory capital concept, not an attempt to reproduce the actual SCR methodology.

It is the underlying daily figure (-2.74%), not the illustrative one-year figure, that is compared with the Extreme Value Theory results in Section 8.

8. Extreme Value Theory: Risk Beyond the Observed Data

All VaR estimates above rely on the 2020–2026 sample and implicitly assume that future extreme losses will not exceed what has already been observed. Extreme Value Theory (EVT) addresses this by modelling only the losses that exceed a high threshold (here, the 95% VaR), fitting a Generalized Pareto Distribution to the 85 resulting exceedances, and using it to extrapolate beyond the observed data — a standard approach in actuarial and reinsurance tail-risk modelling.

The fitted shape parameter ($\xi \approx 0.37$) is positive, providing evidence of substantial tail heaviness in the fitted distribution — consistent with the gap observed between historical and parametric VaR in Section 4. At the 99.5% level, the EVT-based daily estimate (-2.73%) is close to the daily 99.5% historical VaR from Section 7.2 (-2.74%); these are two different calculations on a comparable daily basis, not the same number restated. At the 99.9% level — a point with almost no direct precedent in the historical sample, where a direct empirical estimate would be unreliable — EVT provides a model-based extrapolation of approximately -5.19% , nearly double the 99.5% figure. This is the practical value of EVT in a reinsurance context: it allows reasoning about losses beyond anything in the historical record, rather than assuming the past sets the limit.

9. Limitations & Assumptions

- Constant-weight allocation: portfolio returns are calculated using fixed 60/40 weights applied each day, which is closer to a daily-rebalanced portfolio than a true buy-and-hold position whose weights would drift with market prices. A real institutional portfolio would follow an explicit rebalancing policy.
- Normal distribution assumption: the parametric VaR and Monte Carlo simulation assume normally distributed returns, which understates tail risk relative to the historical and EVT-based estimates.
- Model risk / distributional assumptions: the different VaR and ES estimates in this report depend on the assumed return distribution and tail behaviour (Normal, empirical, or Generalized Pareto). Different distributional or dependence assumptions could produce materially different results.
- ETF proxies, not real portfolios: EUNL.DE and EUN5.DE are simplified proxies; a real institutional portfolio would include a far wider range of instruments, sectors, and currencies.

- No transaction costs or liquidity constraints are modelled.
- Historical sample limitations: the 99% VaR and Expected Shortfall rely on a relatively small number of tail observations, introducing statistical uncertainty despite the Kupiec backtest result.
- Single-day hypothetical shocks scale one worst observed daily return and do not represent a cumulative multi-day event, unlike the historical scenarios.
- The 5.5-year bond duration is an industry-typical estimate, not the fund's officially reported figure.
- EVT threshold choice: results are based on the 95% VaR as the exceedance threshold; a different threshold could yield somewhat different shape and scale estimates, a known sensitivity in Peaks-Over-Threshold modelling.
- The one-year figure in Section 7.2 is a simplified statistical illustration, not a regulatory SCR calculation — the actual Solvency II Standard Formula combines separately shocked risk modules using prescribed correlation matrices, rather than scaling a single VaR estimate by time.

10. Conclusion

This analysis shows that no single risk measure is sufficient to characterise portfolio risk. The historical VaR model passes a formal Kupiec backtest for breach frequency under normal conditions — though this test says nothing about Expected Shortfall or tail severity — and both the historical-vs-parametric gap and the Extreme Value Theory results point to the same conclusion: extreme losses are more likely than a Normal-distribution model would suggest.

Stress testing adds something the statistics alone cannot: it shows concretely that the equity-bond diversification benefit weakened during the 2022 rate-hiking cycle. Adding an interest rate duration shock and an illustrative Solvency II-style capital perspective on top ties the analysis back to how an insurer or reinsurer would think about this kind of portfolio, while being explicit that this is a simplified illustration rather than a full regulatory capital calculation.

None of these methods is sufficient on its own. Used together — statistical, scenario-based, and tail-focused — they give a considerably more realistic picture of portfolio risk than any single measure taken in isolation.